

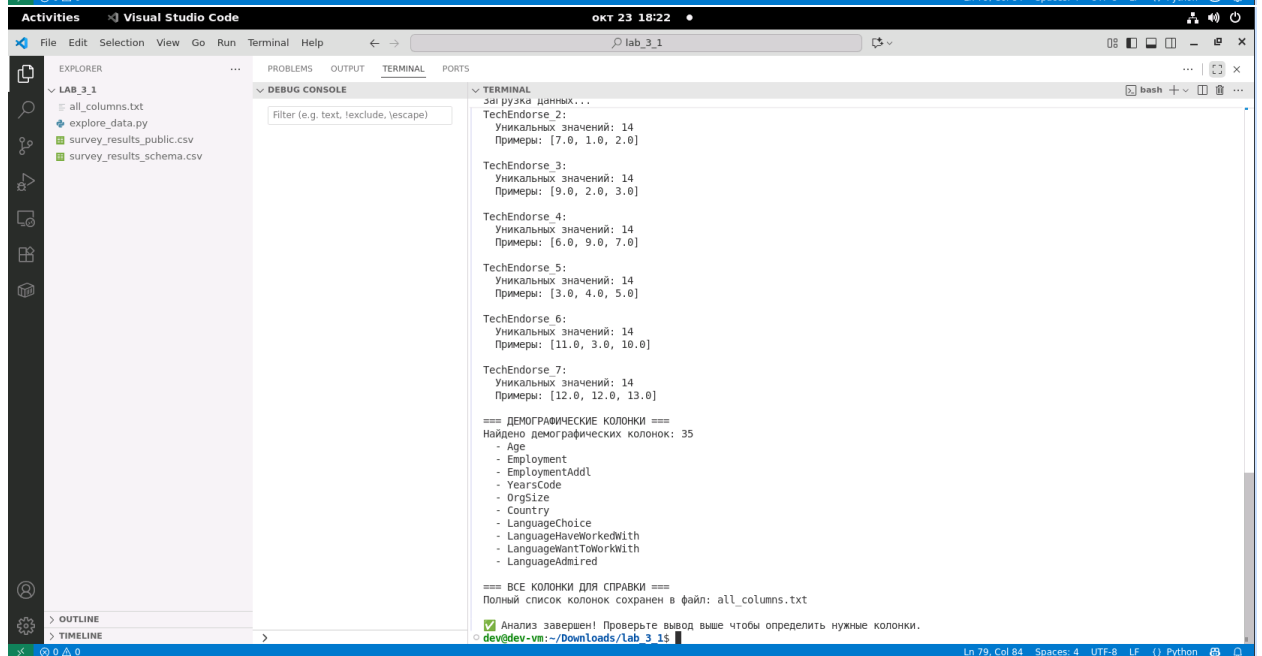
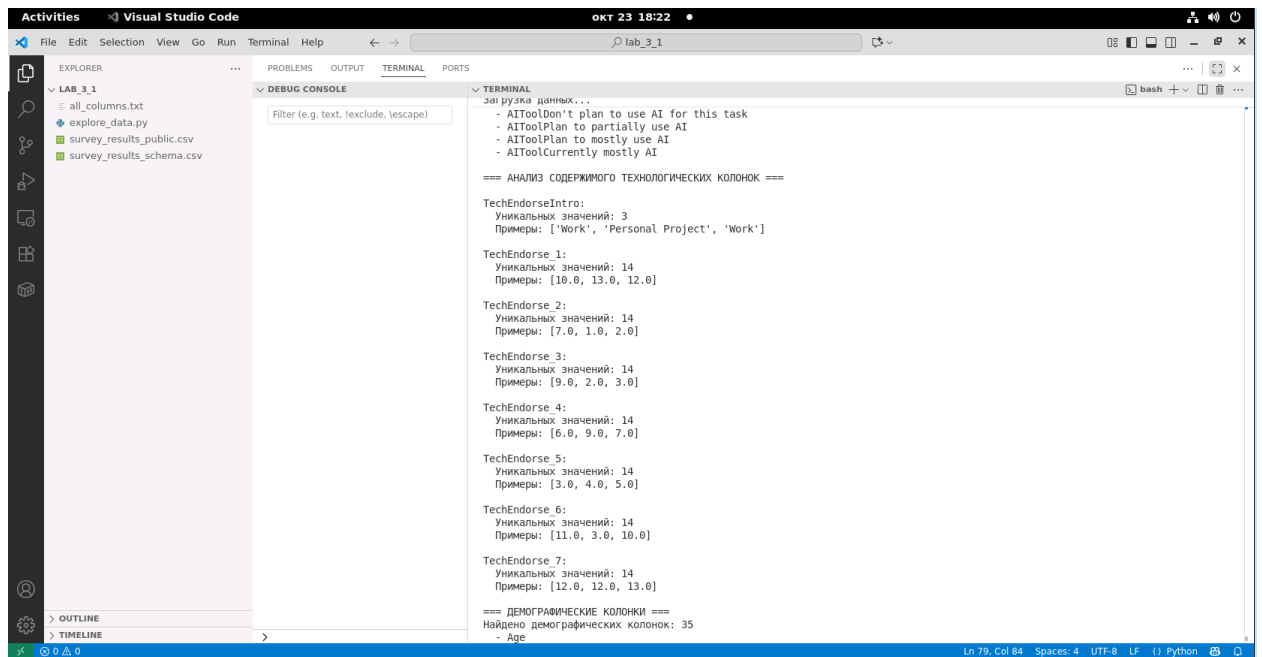
Анализ исходных таблиц с помощью python

The screenshot shows a Jupyter Notebook interface with the following components:

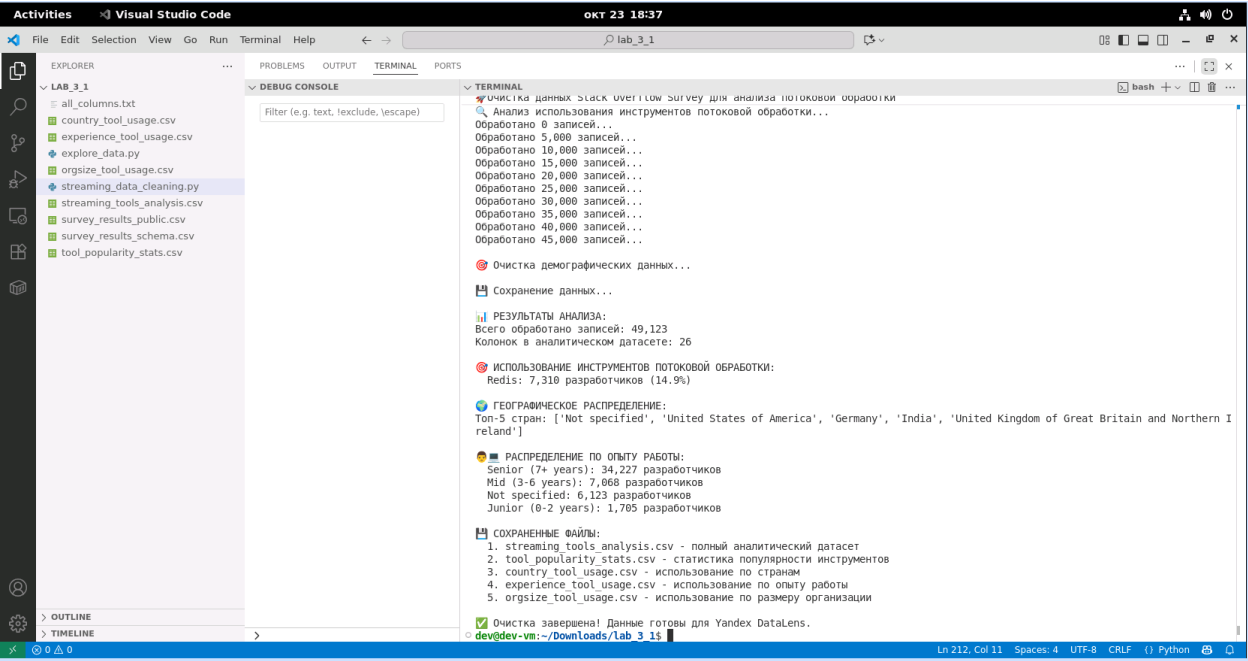
- EXPLORER:** Lists files including `all_columns.txt`, `explore_data.py`, `survey_results_public.csv`, and `survey_results_schema.csv`.
- DEBUG CONSOLE:** Contains a filter input and a list of problems.
- TERMINAL:** Displays the output of the `python3 explore_data.py` command. The output includes:
 - File structure and schema description.
 - Columns in `schema_df`: `['qid', 'qname', 'question', 'type', 'sub', 'sq_id']`.
 - First 5 rows of `schema_df` (rows 0-4).
 - File structure and schema description.
 - Columns in `main_df`: `['ResponseId', 'MainBranch', 'Age', 'EdLevel', 'Employment', 'EmploymentAddl', 'WorkExp', 'LearnCodeChoose', 'LearnCode', 'LearnCodeAI']`.
 - Search for technological columns, finding 53 columns.

The screenshot shows the same Jupyter Notebook interface, but with the following changes:

- EXPLORER:** The file list is updated to include `explore_data.py`.
- DEBUG CONSOLE:** The list of problems is updated to include `all_columns.txt`.
- TERMINAL:** The output of the `python3 explore_data.py` command is updated to show the results of the search for technological columns, including a list of 53 columns.



Очистка данных с помощью python



Дополнительный анализ

☐ ДЕТАЛЬНЫЙ АНАЛИЗ ДАННЫХ

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Размер датасета: (49123, 26)

Колонки: ['Country', 'Age', 'YearsCode', 'Employment', 'OrgSize', 'Apache Kafka', 'Apache Flink', 'Apache Spark', 'Apache Storm', 'Apache Pulsar', 'Amazon Kinesis', 'Google Cloud Pub/Sub', 'Azure Stream Analytics', 'RabbitMQ', 'Redis', 'Apache NiFi', 'Apache Beam', 'Confluent Cloud', 'AWS MSK', 'Google Dataflow', 'Apache Samza', 'Amazon MSK', 'Azure Event Hubs', 'AWS Lambda', 'ExperienceLevel', 'CountryGroup']

☐ СТАТИСТИКА ИНСТРУМЕНТОВ:

Tool Users Percentage

0 Redis 7310 14.88

☐ РАСПРЕДЕЛЕНИЕ ПО СТРАНАМ:

CountryGroup

Not specified 13721

Other 11350

United States of America 7226

Germany 3022

India 2542

United Kingdom of Great Britain and Northern Ireland 2038

France 1409

Canada 1303

Ukraine 964

Poland 887

Name: count, dtype: int64

□ □ РАСПРЕДЕЛЕНИЕ ПО ОПЫТУ:

ExperienceLevel

Senior (7+ years) 34227

Mid (3-6 years) 7068

Not specified 6123

Junior (0-2 years) 1705

Name: count, dtype: int64

□ РАСПРЕДЕЛЕНИЕ ПО РАЗМЕРУ КОМПАНИИ:

OrgSize

Not specified 14979

20 to 99 employees 6680

100 to 499 employees 6104

Less than 20 employees 5597

10,000 or more employees 4959

1,000 to 4,999 employees 4152

500 to 999 employees 2359

Just me - I am a freelancer, sole proprietor, etc. 1760

5,000 to 9,999 employees 1604

I don't know 929

Name: count, dtype: int64

□ ПОИСК ДРУГИХ ИНСТРУМЕНТОВ В ИСХОДНЫХ ДАННЫХ:

□ ТОП-20 БАЗ ДАННЫХ:

PostgreSQL: 13304

MySQL: 9976

SQLite: 9239

Redis: 7279

Microsoft SQL Server: 6781

MongoDB: 5963

MariaDB: 5669

Elasticsearch: 4310

Oracle: 2562

Dynamodb: 2470

BigQuery: 1627

Supabase: 1525

Cloud Firestore: 1439

H2: 1296

Firebase Realtime Database: 1264

Microsoft Access: 1195

Cosmos DB: 1157

Snowflake: 1059
InfluxDB: 945
Databricks SQL: 864

☐ ТОП-15 ПЛАТФОРМ:

Docker: 17162
npm: 13852
Amazon Web Services (AWS): 10270
Pip: 9970
Kubernetes: 6959
Homebrew: 6267
Vite: 6206
Microsoft Azure: 6125
Google Cloud: 5900
Make: 5642
Yarn: 5172
Cloudflare: 4897
NuGet: 4592
Webpack: 4495
APT: 4493

☐ КОМПЛЕКСНЫЙ АНАЛИЗ ИНСТРУМЕНТОВ ПОТОКОВОЙ ОБРАБОТКИ

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☐ ИСПОЛЬЗОВАНИЕ ИНСТРУМЕНТОВ ПОТОКОВОЙ ОБРАБОТКИ:

Redis: 7,310 разработчиков (14.88%)

☐ ИСПОЛЬЗОВАНИЕ ПО ОПЫТУ РАБОТЫ:

Senior (7+ years):

Redis: 6358 (18.6%)

Mid (3-6 years):

Redis: 829 (11.7%)

Not specified:

Redis: 49 (0.8%)

Junior (0-2 years):

Redis: 74 (4.3%)

☐ ИСПОЛЬЗОВАНИЕ ПО СТРАНАМ (ТОП-5):

United States of America:

Redis: 1490 (20.6%)

Germany:

Redis: 564 (18.7%)

India:

Redis: 456 (17.9%)

United Kingdom of Great Britain and Northern Ireland:

Redis: 376 (18.4%)

France:

Redis: 305 (21.6%)

☐ ИСПОЛЬЗОВАНИЕ ПО РАЗМЕРУ КОМПАНИИ:

20 to 99 employees:

Redis: 1408 (21.1%)

500 to 999 employees:

Redis: 439 (18.6%)

10,000 or more employees:

Redis: 700 (14.1%)

Less than 20 employees:

Redis: 1110 (19.8%)

5,000 to 9,999 employees:

Redis: 245 (15.3%)

100 to 499 employees:

Redis: 1265 (20.7%)

1,000 to 4,999 employees:

Redis: 735 (17.7%)

I don't know:

Redis: 83 (8.9%)

Just me - I am a freelancer, sole proprietor, etc.:

Redis: 229 (13.0%)

☐ СОЗДАНИЕ АГРЕГИРОВАННЫХ ДАННЫХ ДЛЯ DATALENS...

✓ СОЗДАНЫ ФАЙЛЫ ДЛЯ DATALENS:

- datalens_tools_summary.csv
- datalens_country_tools.csv
- datalens_experience_tools.csv
- datalens_orgsize_tools.csv

Результат выполнения очистки и предварительного анализа данных

✓ LAB_3_1

- ≡ all_columns.txt
- ⚙ comprehensive_analysis.py
- 📄 country_tool_usage.csv
- 📄 datalens_country_tools.csv
- 📄 datalens_experience_tools.csv
- 📄 datalens_orgsize_tools.csv
- 📄 datalens_tools_summary.csv
- ⚙ detailed_analysis.py
- 📄 experience_tool_usage.csv
- ⚙ explore_data.py
- 📄 orgsize_tool_usage.csv
- ⚙ streaming_data_cleaning.py
- 📄 streaming_tools_analysis.csv
- 📄 survey_results_public.csv
- 📄 survey_results_schema.csv
- 📄 tool_popularity_stats.csv